

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 311 — Past Questions & Materials
by Renance

Course: CHE 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200, 300

This study pack was compiled and restructured by Renance for CHE 311 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — CHE 311 Material and Energy Balance Calculation

Level 300 • First Semester • PastQuestion

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oe t Semester fxamination, 2024/20

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t CALCULATION

SE TITLE: MATERIAL AND KNERGY DAT ry aNty' 3

AoUnee CODE: CHE 311

TIME ALLOWED: 3 Hours

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CTIONS: Answer Any Five Questions Start each Questi

INSTRUC' : '

i onents A, B and C is fed into distillation

l(a) A crude oil consisting of cqual weights of Hap , iitck ee ae

lumn at 10,000kp/hr. It is to be separated into three (3) } . eae

: 'aibieg 90% A, 7% B, and 3% C, a side stream Sonim 20% 'a _ eae he

mis a If he overhead stream is obtained at a rate of 2,800kg/hr and th

: ich it is obtained.

3.500kg/hr, find the composition of the bottom stream and the rate at which it is

(b) It 25000 kg of whole milk containing 4% fat is to be separated in 6 hours period into skim

milk with 0.25 % fat and creara with 25 % fat, what are the flow r

ates of the two output streams

from a continuous centrifuge which accomplishes this separation?

2a) A {eed of 10% brine (N

50.000ke~h. if 25,000kg/h of

the ex}

ACI solution) is to be concentrated in an e
vapour is obtaine
(streams. Draw the schematic diagram.

vaporator, it enters at

d, find the weight percent of the components in

(b} Assume a distillation column is to be

contains 20.0% by mole. N-pa

pentine ¢

| lip = 40,qg0Km0! nh

oo

peer ees

. Te

Kop = 20"8]

xe. 7 1-XeP

a

on column

ow diagram for distillati

xture containing 45% benzene (B) and 55% toluen

am of 95 wt% B is produced and 8 %

The feed rate is 2000kg/hr. Determine:

luene in the bottom stream

Fig, 1.0 Process fl

e (T) by mass is fed to a distillation

of the benzene fed to the column

3(a) A mi

the overhead flow rate and

mn. An overhead stre

he bottom stream.

ow rate of benzene and to

colu

leaves wm

the mass fl

% carbon and 22.3% ash. The

of coal from the mine is formed to contain 67.2

% carbon and the rest is ash.

(b) A sample

residue obtained at the end of combustion is analyzed to contain 7.1

Compute the percent (%) of the original carbon un-burnt in the residue.

usually water, a scrubber is used to remove dust or fine particles from

4(a) Using a liquid spray, the entrance gas stream so that the outlet gas stream satisfies process or emission requirements standard. How much water must be continually fed to wet the scrubber as shown in Fig 2.0 in order to keep the unit running?

Each of the streams is identified by a number located in a diamond symbol. Stream 1 is recycled

scrubber and it is 5.54 m³/h. The liquid being withdrawn for treatment and disposal (stream 4) is 0.545 m³/h.

View 2.0P

weight) to produce an output containing 30% solids
100 kg of feed to the evaporator

(b) A solution which is 80% oil, 15% unusable by-product and 5% impurities enters a refinery. One output is 92% oil and 6% unusable by-product. The other output is 60% oil and flows at a rate of 1000 L/hr. (Assuming no accumulation).

(i) What is the flow rate of the input stream?

(ii) What is the percent composition of the 1000 L/hr output stream?

(iii) What percent of the original impurities are in the 1000 L/hr output?

y. How much

powder of 80% complete

enters and exits continuous!

6(a) Imagine a continuous process where full milk &
45% total solid concentrated whole milk is needed to make
our, Draw the schematic diagram

make 200 kg/h of p

solid milk powder each h
which contains 25%

ed with another solution wh
ters at 50 kg/h,

solids 18 MIX
solution en

(b) A solution which contains 10%
solid is removed. If the 10%

A single output which is 20%

solid,

flow rates Assuming no ac

What are the other flow

modified air of 0.006 kg H₂O/kg air is

cumulation?

er. The atmospheric

required to be fed to a dry

ugh a

is passed thro

Tai) Dehu

air has the absolute humidity of 0.02 kg H₂O/kg air. Therefore, the air

m that reduces the absolute humidity to 0.004 kg H₂O/kg air. 5°; part of the

mass of water removed per }06

dchumidifying syste

{resh atmospheric air is bypasse

kg of dry air feed and the percent

d to dechumidifying unit. Find the

age of bypassed air. Draw the schematic diagram

{ances as shown in the table below.

(ii) Consider the enthalpies of fission for various subs

'Table 1.0 ; a

Substance | melting point (°C)

Subsiane

oe

Water

-114.3

Aluminum

Benzene

Ethanol

e when 40.7g of water melt at 0°C

(i) Determine the energy chang

(ii) Determine the energy change when 106g of benzene freezes at 5.5°C

eralized water, a shel! and tube heat ex-changer is cooled. Evaluate the to!

cal/hr), if the water flow rate is 400m³/hr and temperature rise

(b) With a demin

ling, watcr (is

heal rejected to coo

9°C.

Paper 2 of 2 — CHE 311 - 2024/2025 - First Semester - MATERIAL AND ENERGY BALANCE CALCULATIONS

Level 200 • First Semester • PastQuestion

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ENGINEERING AND eCHNOLOGY a

COLLEGE OF :

DEPARTMENT OF CHEMICAL NGINEERING

ination, 2024/2025 Academic Session

Q CALCULATIONS

First Semester exami

SE TITLE: MATERIAL AND ENERGY BALANCE

COUR CREDIT UNITS: 3

COURSE CODE: CHIE 311

TIME ALLOWED: 3 flours sie

INSTRUCTIONS: Answer Any Five Questions Start each Question on a New Page-

s A, B and C is fed into distillation

de oil consisting of equa

00kg/hr. It is to 9¢ separated into three (3) fractions namely, an overhead stream

IY Ws and 3% C, a side stream containing 20% A, 70% BA0KC and a

he overhead stream ;s obtained at a rate of 2,800ke/hr and the side stream ah

tion of the bottom stream. and the rate at which it is obtained. F

(a) A cru 1 weights of component

column at 10,0

containing 90% A

bottom stream. if t

3.500k¢/hr, find the composi

skim

ted in 6 hours period into

fat is to be separa

the two output streams

containing 4%

e flow rates of

(bp It 25000 kg of whole milk

milk with 0.25 % fat and crear with 25 % fat, what are th

a continuous centrifuge which accomplishes this separation?

0% brine (NaCl solution) is to be

g/h of vapour is obtained, find

the schematic diagram,

from

it enters at

concentrated in an evaporator,

25,000 kg/h. If 25,000 kg/h is the weight percent of the components in

the exit streams. Draw the

to separate a 10,000 kmol/h feed stream that

N-hexane into streams that are 85% pure

the, find the output

distillation column is to be used

and the remaining

balance, assuming steady state

(b) Assume a distillation

contains 20.0% by mole. N-pentane and

isopentane and 85% pure hexane. Using mass

streams flow rates, See figure 1.0.

1.0

Figure 1.0

Kap* BSty

Kons t- Aer

Xen Sts Xep

Figure 1.0

Np =? ae

Se 8

Xep1-Xen Bs a yon

Xp = 95%

Figure 1.0 Process flow diagram for distillation column

See et Sata Resht ears Bir Shs, le 'eet SRS Mae tange iii.' yi liac

Figure 2.0 Process operation of material balance

pricy wha you understand by Plug Flow Reactors (PER;) and Fluidized Bed

concentrates a liquor containing solids 6% by w/w (weight by

weight) 30% solids w/w. Calculate the evaporation of water per

kg of unusable by product and 5% impurities enters a refinery.

the by-product. The other output is 60% oil and flows at a

distillation).

Le}h

inuously. How much

: 5 N its cont Be

6(a) Imagine a continuous process where full milk enters and exit er of 80% complete

45% total solid concentrated whole milk is needed to make 200 ke/h of pow

solid autk powder each hour, Draw the schematic diagram

; i 25%

(b) A solution which contains 10% solids is mixed with another solution which hee ee solid. A single output which is 20% solid is removed. If the 10% solution enters at 3. Eee What are the other flow rates Assuming no accumulation? sete

ai) Dehumiditied air of 0.006 kg HbO/kg air is required to be fed to a dryer. The atmosphere : hoi air has the absolute humidity of 0.02 kg H2O/kg air, Therefore, the air is passed throughl he es dehumidilying system that reduces the absolute humidity to 0.004 kg H2O0/kg air. So, part of the —

‘ fresh atmospheric air is bypassed to dehumidifying unit. Find the mass of water removed per 100 eee Ag of dry air feed and the percentage of bypassed air. Draw Me the schematic diagram Fe,

(ii) Consider the enthalpies of fission for various substances as shown in the table below.

‘Table 1.0

muetting point SS) AB tus (kJ/mol)

: 6.0 She

1 | 660_ ae ee

| = és 9.95

“a 5.02